OPX1000

A control platform for large-scale quantum computers

Highlights:

Trusted by builders of the world's largest and most advanced quantum computers, OPX1000 is a best-in-class control platform for a wide span of qubit types.

OPX1000 features unparalleled processing resources and channel density and is easily expandable to cost-efficiently control even the largest quantum processors available.

- Processor-based quantum controller™
- Highest density in the industry – 80 analog channels / 3 U
- Cutting-edge analog frontend, with DDS up to 10.5 GHz
- Intuitive QUA coding – high-level pulse programming language.

Scale up your quantum computer while enhancing flexibility and performance with OPX1000.

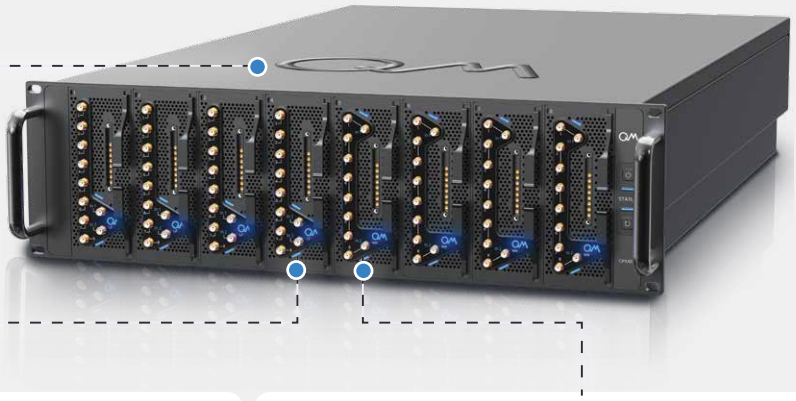
Key Benefits:

- **Agile and flexible:** Robust real-time computation, real-time control-flow, and ultra-fast parametric feedback (not just Boolean) with active reset < 160 ns.
- **Smooth scalability:** Effortlessly add units with any to-any data-sharing, exceptional phase synchronization between all channels, and no software redesign.
- **Unmatched fidelities:** State-of-the-art analog specs, ultra-fast calibrations, and re-tuning in real-time.
- **Shortest time to results:** Easy to set up (RB in less than 48 hours from unboxing), intuitive programming, and fastest runtime.
- **Long-term operation:** Hot-swappable critical parts, CE and FCC safety certifications.
- **HPC and data-center ready:** Native GPU hardware integrations, multi-users, out-of-the-box workflows, smooth integration with upper-level software environments.

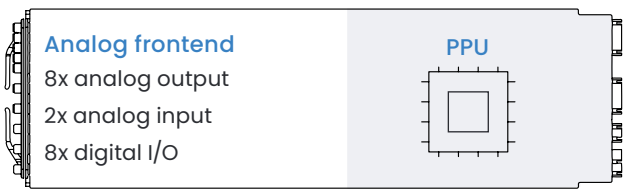
Modular Design

High-Density 3U Chassis

Up to 8 modules of two types may be used per chassis, in any combination

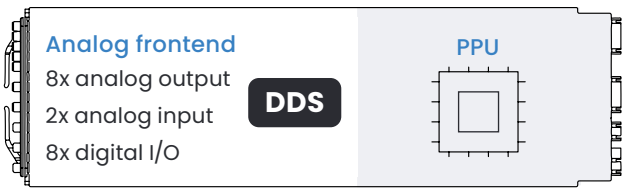


Low-Frequency Module (LF-FEM)



- DC - 750 MHz
- 16-core Pulse Processing Unit (PPU), proprietary
- Up to 16 tones per channel

Microwave Module (MW-FEM)



- Direct Digital Synthesis (DDS) technology up to 10.5 GHz
- 16-core Pulse Processing Unit (PPU), proprietary
- Up to 8 tones around 2 carriers per channel, i.e., 16 carriers per FEM

OPX1000 Specs

Chassis General	
Height	3U
Maximum capacity	8 low-frequency and microwave modules, in any combination
Mounting options 19" rack and desktop	19" rack and desktop
Management ports	1 : 2 (2x 1 GbE, 1x 100 GbE)
Clock input ports	1 : 1, (10 MHz, 2 GHz more options per request)
Clock output ports	4
Fans	5 + 1, hot swap
Power supplies	2 + 1, hot swap (100 - 240 VAC, 50 - 60 Hz)

Front-End Module (FEM) General	
FPGA	UltraScale+TM XCVU13P
Pulse Processing Unit (PPU) cores	16, parallel operation, all-to-all connectivity
Channels	8 analog output / 2 analog input / 8 digital I/O
Analog channels resolution	16 bits DAC / 12 bits ADC
Analog channels sampling rate	LF-FEM: 2 GSa/s MW-FEM: 2 GSa/s (1 GSa/s per I and Q, normal mode)

Microwave FEM Analog Output Channels	
Frequency range	50 MHz - 10.5 GHz
Instantaneous bandwidth	800 MHz
Power	Up to +10 dBm
Maximum tones per channel	8
Phase noise @ 6 GHz with 10 kHz offset	< -125 dBc/Hz
Jitter @ 6 GHz, ∫(1 Hz - 100 MHZ)	External clock: < 150 fs
SFDR across all supported frequency range	> 60 dBc
Total harmonic distortion (THD) outside bandwidth	> 40 dBc
Signal to noise ratio @ +10 dBm	150 dBc

Microwave FEM Analog Input Channels	
Frequency range	1.8 - 10.5 GHz
Power	-40 dBm to -10 dBm

Low-Frequency FEM Analog Output channels	
Maximum tones per channel	16
Jitter (100 Hz - 100 MHz)	Internal clock: < 350 fs External clock: < 250 fs
Crosstalk channels isolation	> 80 dB @ 100 MHz > 70 dB @ 500 MHz
Crosstalk compensation matrix Built-in	Built-in

Low-Frequency FEM Analog Output Channels Cont.		
FIR/IIR distortion compensation filters	Built-in	
Total harmonic distortion	> 65 dBc	
SFDR (direct mode)	> 60 dBc @ DC - 700 MHz > 55 dBc @ 700 - 750 MHz	
Controllable skew adjustment resolution	10 ps	
Output Mode	**Direct**	**Amplified**
Voltage	1 Vpp (50 Ω load)	5 Vpp (50 Ω load)
Bandwidth	750 MHz	330 MHz
Rise time	< 500 ps @ 0.8 V step	< 900 ps @ 5 V step < 600 ps @ 0.5 V step
Voltage Noise (PSD)	< 1 μV/ √Hz @ 1 Hz < 13 nV/ √Hz @ 10 kHz < 10 nV/ √Hz @ 100 kHz < 6 nV/ √Hz @ 1 MHz < 4 nV/ √Hz @ 10 MHz	< 3.5 μV/ √Hz @ 1 Hz < 40 nV/ √Hz @ 10 kHz < 30 nV/ √Hz @ 100 kHz < 20 nV/ √Hz @ 1 MHz < 15 nV/ √Hz @ 10 MHz
Voltage Noise (RMS)	< 100 μVrms	< 300 μVrms

Low-Frequency FEM Analog Input Channels	
Bandwidth	750 MHz
Controllable gain	32 dB
Time tagging resolution	< 50 ps

Front-End Module (FEM) Digital I/O	
Digital I/O channels sampling rate	1 GSa/s
Voltage	3.3 V (LVTTL) / 5 V (TTL)
Output rise time	< 0.4 ns
Controllable skew adjustment resolution	1 ns

System & Real-Time	
Minimal feedback latency - analog input to digital output	< 160 ns
Minimal feedback latency - analog to analog (LF FEM)	< 200 ns within FEM < 300 ns within OPX1000 < 500 ns between OPX1000
Minimal feedback latency - analog to analog (MW FEM)	< 160 ns within FEM < 200 ns within OPX1000 < 400 ns between OPX1000
Between modules added latency	100 ns intra-chassis 300 ns inter-chassis
Operation latencies	Amplitude update < 16 ns Phase update < 32 ns Frequency update < 150 ns

About Quantum Machines

Quantum Machines (QM) accelerates the realization of practical quantum computing that will disrupt all industries. Our comprehensive portfolio includes state-of-the-art control and cryogenic electronic solutions that support a wide span of qubit technologies. QM's OPX family of processor-based quantum controllers™ delivers unprecedented performance, scalability, and productivity. OPX controllers are extremely agile and flexible, capable of running even the most complex quantum algorithms right out of the box, such as quantum error correction, multi-qubit calibration, mid-circuit frequency tracking, and more. With hundreds of deployments, Quantum Machines' solutions have been widely adopted by research labs, HPC centers, full-stack quantum computer manufacturers, and cloud service providers. For more information, please visit - quantum-machines.co

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